CLAIMPROMOTEBENCH: SYNTHETIC POLICY-CONSTRAINED EVALUATION OF EVIDENCE-STATE TRANSITIONS

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ABSTRACT

Producing evidence and deciding what status that evidence licenses are different tasks. We frame the latter as synthetic policy conformance, rather than a broad test of research-agent ability. CLAIMPROMOTEBENCH presents a claim, typed evidence, a current state, and a requested transition under a fixed author-defined policy. In a preregistered 300-item hidden study, a Qwen3.7-Max service configuration returned 272 valid decisions (decision coverage .9067) and 123 complete pairs (pair coverage .8200). End-to-end item and pair accuracy were .8600 and .7533; conditional accuracy among valid decisions and complete pairs was .9485 and .9187. End-to-end recall was .9533/ALLOW, .8209/HOLD, 0/DEMOTE, and .6250/ESCALATE. A weak public-development text Naive Bayes comparator reached .6133 item and .2267 pair accuracy. A separately precommitted public robustness challenge balanced 90 ALLOW members against 30 each of HOLD, DEMOTE, and ESCALATE over 90 pairs and 15 factors. Qwen produced 148/180 valid decisions, with .7333/.5333 end-to-end and .8919/.7500 conditional item/pair accuracy; a Flash route produced 146/180 valid decisions, with .5000/.1444 end-to-end and .6164/.2131 conditional accuracy. Every service, syntax, schema, and semantic failure remains incorrect in the end-to-end denominator. The earlier Flash lane's 300 HTTP 401 responses diagnose its frozen service interface; the later route establishes executability only. Different service stacks preclude a strict causal model ablation. A deterministic policy executor is a definitional policy ceiling, not a learned baseline.

1 INTRODUCTION

A valid artifact licenses only the scope that was actually checked. A finite computation can support its registered finite domain; a proof certificate can support its exact encoded statement in a pinned environment; and an internal reproduction carries different provenance from an independent review. A system that conflates these evidence types can assign a claim status stronger than the evidence state warrants.

We study the narrow decision between evidence production and claim-state assignment:

> *Given a claim, typed evidence, a current state, and a requested transition, which action conforms to a fixed, machine-checkable policy?*

This is a synthetic policy-conformance problem. It does not measure open-ended discovery, experimental design, literature synthesis, or general research-agent competence. The policy is an authored operational definition, so oracle agreement measures behavior relative to that definition rather than scientific truth or community consensus.

CLAIMPROMOTEBENCH represents research status, computational support, formal certification, and independent review as separate state axes. Controlled pairs keep the claim and target transition fixed while changing one registered prerequisite. We score four policy actions—ALLOW, HOLD, DEMOTE, and ESCALATE—and retain unusable responses in the end-to-end denominator. This design exposes three distinct questions: whether a service returns a valid decision, whether the decision is correct when valid, and whether both members of an evidence contrast are handled correctly.

Our contributions are:

1. a synthetic, policy-constrained evaluation of evidence-state transitions with executable numerical and formal checks and local controlled contrasts;
2. failure-aware item and pair estimands that report coverage, conditional correctness, end-to-end correctness, action recall, and a no-valid-decision confusion column;
3. two complementary evaluations: a preregistered hidden realization of the 20-factor benchmark and a precommitted public robustness challenge balanced across all three non-ALLOW actions; and
4. a replayable evidence chain that separates model behavior, service-interface behavior, a definitional policy ceiling, and privacy-filtered reviewer artifacts.

2 Related Work

Behavioral and contrastive evaluation. CheckList organizes behavioral tests around capabilities and expected invariances (Ribeiro et al., 2020), while contrast sets probe local decision boundaries through small, meaningful input changes (Gardner et al., 2020). Metamorphic testing similarly checks relations among transformed inputs when a conventional test oracle is difficult to apply (Chan et al., 1998). ClaimPromoteBench uses authored policy labels and controlled pairs: most pairs intentionally change the action, while separate metamorphic checks preserve the action under identifier renaming, evidence reordering, and irrelevant-evidence insertion. These tests characterize selected local behaviors; they are not an estimate of natural-distribution performance.

Structured output and selective prediction. Incremental parsing and grammar-constrained decoding can prevent structurally invalid generations (Scholak et al., 2021; Geng et al., 2023). They cannot guarantee that a structurally valid action satisfies cross-field semantics or the evidence policy. Selective classification studies risk–coverage trade-offs and reject options (Geifman & El-Yaniv, 2017; 2019); recent abstention benchmarks expose continuing failures on unanswerable tasks (Kirichenko et al., 2025). We therefore report coverage and conditional correctness together, while end-to-end metrics count every absent or invalid decision as incorrect. HOLD is a policy action about current evidence, whereas service failure is recorded as no valid decision.

Scientific evidence and research-agent evaluation. CLAIM-BENCH, DeepFact, and ClaimDB evaluate scientific claim extraction or factual verification (Javaji et al., 2025; Huang et al., 2026; Theologitis et al., 2026). ScienceAgentBench, CORE-Bench, and POPPER evaluate executable research tasks, computational reproduction, or hypothesis falsification (Chen et al., 2025; Siegel et al., 2025; Huang et al., 2025). SeekBench, Omni-Decision, EviGraph, and ClaimLedger emphasize grounding or explicit evidence maintenance (Shao et al., 2026; Ma et al., 2026; Ren et al., 2026; Elliman, 2026). ClaimPromoteBench complements these tasks by fixing an authored policy and isolating the status-transition decision after evidence has been registered. Formal artifacts are interpreted only for their exact statements, consistent with the narrow verification scope highlighted by OSVBench (Li et al., 2026).

3 Task and Evidence Flow

Each item is $x = (c, E, s, \tau, t)$: a claim c, typed evidence E, current state s, requested target τ, and frozen evaluation time t. The state has four axes:

$$\begin{aligned} \texttt{research_status} &\in \{\texttt{PROPOSED}, \texttt{OPEN}, \texttt{RESOLVED}\}, \\ \texttt{computation_status} &\in \{\texttt{NONE}, \texttt{COMPUTATIONAL}\}, \\ \texttt{formal_status} &\in \{\texttt{NONE}, \texttt{FORMALLY_CERTIFIED}\}, \\ \texttt{review_status} &\in \{\texttt{UNREVIEWED}, \texttt{INDEPENDENTLY_VERIFIED}\}. \end{aligned}$$

The public policy checks dependencies, scope, freshness, provenance, runtime completion, contradictions, statement identity, and artifact integrity. Its deterministic oracle $\pi_P(x)$ returns one of four actions: ALLOW, HOLD, DEMOTE, or ESCALATE. ALLOW accepts the requested target; HOLD preserves the current state because support is insufficient; DEMOTE weakens a state whose recorded

basis has failed; and ESCALATE preserves the state while flagging an unresolved authoritative contradiction.

Let $v(x)$ indicate a syntactically, schema-, and semantically valid response and let $\hat{y}(x)$ be its action. End-to-end item correctness is

$$C_{\text{item}}(x) = \mathbb{1}[v(x) = 1]\mathbb{1}[\hat{y}(x) = \pi_P(x)].$$

For controlled pair $i = (x_i^a, x_i^b)$, $C_{\text{pair}}(i) = C_{\text{item}}(x_i^a)C_{\text{item}}(x_i^b)$. Decision coverage is the fraction with $v = 1$; pair coverage is the fraction with both members valid. Conditional accuracies divide correct items or pairs by those coverage denominators. End-to-end accuracies divide by all scheduled items or pairs.

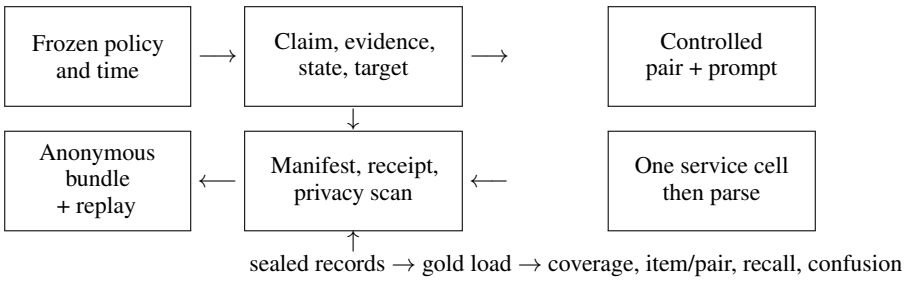


Figure 1: Benchmark and evidence flow. Model-facing inputs are fixed before service calls. End-to-end analysis retains every terminal failure; reviewer artifacts are derived only after records are sealed and privacy checked.

Within a controlled pair, only registered fields relevant to one factor change. For example, a certificate with a missing dependency maps to HOLD; adding that exact dependency maps to ALLOW. Separate invariance tests rename identifiers, reorder evidence, or add irrelevant records and require both the oracle and rendered prompt semantics to remain unchanged.

4 Benchmark and Evaluation Design

4.1 Synthetic benchmark

Version 1 contains 200 pairs (400 items) over 20 factors spanning dependencies, numerical sufficiency, runtime, freshness and contradiction, provenance and review, and formal identity and trust base. Fifty pairs form public development data; 150 pairs form the hidden evaluation. Numerical cases use exact-rational finite-grid artifacts whose registered values and scope are recomputed. Formal cases use generated Lean sources under a pinned toolchain. These checks establish internal policy consistency, not external scientific validity.

The fresh r6 realization was constructed after earlier outcomes under a new public commitment, private salt, future randomness beacon, construction receipt, and frozen request schedule. It contains 150 hidden pairs and 300 unique prompts from the same ontology and mostly the same template family. Its gold distribution is imbalanced: 150 ALLOW, 134 HOLD, eight DEMOTE, and eight ESCALATE items. All 600 terminal cells for two registered lanes were sealed before gold loading and analysis. Each item received one call with zero client retries. A text Naive Bayes classifier trained only on the 100 public-development items was frozen as a low-cost comparator.

4.2 Balanced public robustness challenge

To expose action asymmetry, a separately precommitted public challenge selects 15 factors and six test pairs per factor. Its 90 pairs contain 90 ALLOW members and exactly 30 members for each of HOLD, DEMOTE, and ESCALATE. The complete dataset, labels, prompts, adapters, analysis, and execution contract were frozen before inference. Each of two service routes made exactly 180 calls with zero client retries. This is a public, preregistered robustness challenge: it tests behavior on a known policy and public synthetic construction, not generalization to a hidden or natural distribution.

The two routes differ in model and service stack, including output-constraining behavior. Their results are descriptive service-configuration outcomes. They do not identify a causal effect of model weights, provider, schema mode, or any single component.

4.3 NATURALISTIC-CANDIDATE DECISION

A deterministic pointer-only sample of 12 CORE-Bench training candidates was frozen before any CLAIMPROMOTEBENCH model call for this question. Source-level review found task prompts and benchmark answer fields, but no candidate record supplied the time-stamped execution artifact, dependency chain, runtime status, and provenance needed to map a transition under the fixed policy. Capsule-level redistribution terms also require case-specific review. We therefore created zero naturalistic benchmark cases and performed zero model calls. No human annotators, human gold labels, or agreement statistic are claimed. Naturalistic transfer remains open.

4.4 SECONDARY DIAGNOSTICS

Earlier studies are retained as diagnostics rather than a chronology of model development. The original 7,200-decision v1 hidden matrix produced 5,327 format errors; all 1,873 parseable raw decisions selected `ALLOW`, yielding .1333 end-to-end item accuracy and zero pair accuracy. A post-v1 public formatting control produced 108 valid decisions among 160; schema-constrained runs accounted for 78/80 valid decisions and 38/40 complete pairs, yet every valid action remained `ALLOW` and pair accuracy remained zero. An adaptive hidden service study and a one-pass GPT-5.4 public-development follow-up show that outcomes depend strongly on service qualification and configuration. They are descriptive and do not alter the two primary evidence regimes above.

5 RESULTS

5.1 COVERAGE, CONDITIONAL ACCURACY, AND END-TO-END ACCURACY

Table 1 reports both estimands. The r6 Qwen configuration returned 272/300 valid decisions and 123/150 complete pairs. Its .9485 conditional item accuracy becomes .8600 end to end; its .9187 conditional pair accuracy becomes .7533 end to end. Fourteen HTTP 500 responses, one HTTP 503 response, ten timeouts, and three semantic-invalid outputs remain incorrect. Against the frozen Naive Bayes comparator, Qwen added 74 correct items; across the 94 pair blocks with unequal correctness it won 84 and lost 10 (one-sided exact $p = 5.16 \times 10^{-16}$, two-hypothesis Holm-adjusted $p = 1.03 \times 10^{-15}$). The comparator's .6133 item and .2267 pair accuracy confirm that the released text baseline is weak rather than a competitive estimate of the best shallow method.

Table 1: Coverage and accuracy. Conditional values use only valid decisions or complete pairs; end-to-end (E2E) values retain all scheduled failures. "Old" identifies the frozen r6 interface. N/E means model efficacy is non-evaluable because no model output was returned.

Configuration	Valid	E2E item	Cond. item	Pair cov.	E2E pair	Cond. pair
r6 Qwen3.7-Max	272/300	.8600	.9485	.8200	.7533	.9187
r6 public-dev text NB	300/300	.6133	.6133	1.0000	.2267	.2267
r6 old Flash interface	0/300	N/E	N/E	.0000	N/E	N/E
Public challenge Qwen	148/180	.7333	.8919	.7111	.5333	.7500
Public challenge Flash	146/180	.5000	.6164	.6778	.1444	.2131

The r6 Flash configuration returned 300 HTTP 401 responses and no model output. This was a failure of the old frozen service interface, so model efficacy and the registered second confirmation are non-evaluable. In the later public challenge, the new Flash route returned 146 valid decisions. That establishes that the newer service path was executable under its challenge contract; it does not retrospectively repair r6 or support a strict causal comparison.

On the balanced challenge, Qwen recorded 31 transport failures and one semantic failure; Flash recorded 34 semantic failures. Qwen achieved 132/180 correct items and 48/90 correct pairs. Flash achieved 90/180 correct items and 13/90 correct pairs. The conditional scores are materially higher

than the corresponding end-to-end scores, especially for pairs, showing why both coverage and correctness must be retained.

5.2 Action recall and full confusion matrices

Table 2 shows end-to-end recall, so no-valid-decision cells count as misses. R6's eight-item DEMOTE and ESCALATE supports are too small for a uniform four-action claim. The balanced challenge supplies 30 invalid members per action and preserves the same qualitative DEMOTE weakness: Qwen recovers 4/30 and Flash 0/30.

Table 2: End-to-end recall by gold action; no-valid cells are misses. Gold supports by row: r6 150/134/8/8; each challenge lane 90/30/30/30.

Configuration	ALLOW	HOLD	DEMOTE	ESCALATE	Bal. recall	Macro-F1
r6 Qwen3.7-Max	.9533	.8209	.0000	.6250	.5998	.6265
Public challenge Qwen	.8778	.8667	.1333	.7667	.6611	.7004
Public challenge Flash	.7556	.3000	.0000	.4333	.3722	.3928

Macro-F1 is a fixed-four-class mean with zero division set to zero. A post-analysis audit found that the historical challenge aggregate had instead omitted Flash's zero-prediction DEMOTE class, emitting .5237 end to end and .6243 conditionally. The versioned correction receipt binds the unchanged terminal records and historical aggregate; deterministic recomputation gives .3928 and .4682, respectively. No model calls, confusion cells, recalls, or primary item/pair endpoints changed.

Table 3 gives every end-to-end confusion cell. The final column distinguishes absence of a valid decision from any policy action. R6 Qwen maps all eight gold DEMOTE items to ALLOW. On the public challenge, Qwen's 20 valid gold-DEMOTE responses split across all four actions, while Flash never predicts DEMOTE. Flash also maps 13 gold-HOLD items to ALLOW and loses 14/30 gold-ESCALATE items to invalid responses.

Table 3: Full confusion matrices. Rows are gold actions; action columns are valid predictions; "No valid" includes service, syntax, schema, and semantic failures.

Configuration	Gold	ALLOW	HOLD	DEMOTE	ESCALATE	No valid
r6 Qwen	ALLOW	143	0	0	0	7
	HOLD	4	110	0	2	18
	DEMOTE	8	0	0	0	0
	ESCALATE	0	0	0	5	3
Challenge Qwen	ALLOW	79	0	0	0	11
	HOLD	0	26	0	0	4
	DEMOTE	8	4	4	4	10
	ESCALATE	0	0	0	23	7
Challenge Flash	ALLOW	68	4	0	8	10
	HOLD	13	9	0	0	8
	DEMOTE	25	3	0	0	2
	ESCALATE	0	3	0	13	14

5.3 Definitional policy ceiling

The deterministic policy executor reads the same model-visible item and public policy that define the labels. It reproduces 300/300 r6 items and 150/150 pairs, and all 360 scheduled public-challenge item cells and 180 pair cells. This perfect agreement is true by definition of the evaluation target. We call it a *definitional policy ceiling*; it is neither a learned baseline nor independent evidence that the authored policy is scientifically correct. Post-generation gates driven by the same executor are implementation checks and provide no evidence of learned model competence.

6 DISCUSSION

What the positive result supports. The preregistered r6 result supports one time-stamped Qwen service configuration on a fresh hidden realization of the same synthetic ontology. Its .8600/.7533 end-to-end item/pair accuracy and .9067/.8200 coverage show substantial policy conformance under that contract. The higher .9485/.9187 conditional values answer a different question—correctness after a usable response exists. Neither estimand measures broad research ability or naturalistic scientific judgment.

Action balance changes the picture. R6's aggregate is dominated by `ALLOW` and `HOLD`. Its 0/8 `DEMOTE` recall already blocks a uniform competence claim, and the public challenge makes the weakness harder to dismiss: Qwen reaches only 4/30 and Flash 0/30 on `DEMOTE`. The complete matrices also show different error modes. Qwen's challenge errors concentrate in missing decisions and confused demotions; Flash frequently substitutes `ALLOW` for `HOLD` or `DEMOTE` and has lower pair sensitivity. Balanced recall and pair accuracy therefore add information that item accuracy alone hides.

Service reliability is part of end-to-end behavior. Every transport, syntax, schema, and semantic failure is a failed scheduled decision. Conditional reporting remains useful for separating interface reliability from policy reasoning, but it cannot replace the end-to-end estimand. The old r6 Flash 401 result diagnoses a frozen interface that never reached model output. The later Flash route proves only that a different service path executed. Because the routes differ in both service stack and model configuration, the observed Qwen–Flash gap is descriptive and cannot be interpreted as a strict causal ablation.

Comparators and enforcement. The public-development Naive Bayes model gives a reproducible low-cost reference, but .6133 item accuracy, .2267 pair accuracy, .3284 balanced recall, and .3124 macro-F1 make it a weak baseline. Stronger learned comparators remain needed. Conversely, the deterministic executor is maximally strong only because it is the operational definition of gold. Treating it as empirical model evidence would be circular.

7 LIMITATIONS

The benchmark is synthetic and policy-constrained. R6 is a new seeded realization of the same ontology and mostly the same template family, so its hidden design controls leakage more strongly than distribution shift. The balanced challenge is public and precommitted; its 90 pairs and 15 factors measure robustness to an action-balanced synthetic construction, not hidden-test generalization.

Coverage remains a material limitation. R6 loses 28/300 decisions and 27/150 complete pairs; the public challenge loses 32/180 Qwen decisions and 34/180 Flash decisions. Conditional accuracies of .9485/.9187, .8919/.7500, and .6164/.2131 must therefore be read beside their end-to-end counterparts. The confusion matrices reveal severe `DEMOTE` weakness even after balancing, and 30 examples per invalid action still support only coarse action-level estimates.

The evidence covers a few time-stamped service configurations and one call per item. Zero retries make the estimand clear but do not characterize retry policies, sampling variance, fixed model weights, or alternate providers. The old and new Flash routes differ enough that executability of the latter cannot identify why the former returned 401 or what r6 model accuracy would have been. No strict causal provider, model, or structured-output comparison follows.

The policy and labels are authored operational definitions. There is no human annotation study, external policy consensus, inter-annotator agreement, or validation against downstream scientific outcomes. Twelve frozen CORE-Bench candidates lacked the source-level evidence snapshot needed for a defensible mapping, so no naturalistic cases were created or scored. The Naive Bayes comparator is shallow, while the deterministic executor is circular by design. Naturalistic transfer, stronger learned baselines, human-policy agreement, and downstream utility remain open.

8 Reproducibility and Artifact Policy

The original hidden set and exact prompts were fixed before scored inference. Artifact-only recovery reconstructed the same dataset and prompt identity and recomputed the 7,200-record v1 aggregate from all 24 archived runs. It issued no model query and reclassified exactly 27 cross-field-invalid records as format errors by invoking the registered semantic validator omitted by the original runner; no legal model decision was rewritten.

For the fresh study, public C0 and private-salt commitments precede a future randomness beacon and C1 construction receipt. A frozen schedule contains 600 requests, and C2 seals all 600 terminal cells before gold loading or analysis. A post-run audit found cleartext provider hidden-thinking blocks inside 275 response envelopes. Scoring used final response text and parsed decisions alone, so the defect left the C2 order and scientific values unchanged. The public release is a post-C2 text-only allowlist derivative with source-to-derived row and envelope hashes and byte-identical scientific analysis. Its manifest records the generated tree's current file count.

The public balanced challenge froze its dataset, action distribution, prompt packet, adapters, statistics, and one-call zero-retry schedule before inference. Its receipt validates all 360 item-level records and reports zero provider-account findings, retained provider response identifiers, and hidden-reasoning findings. The challenge is public robustness evidence and is not a hidden-test generalization estimate or a causal comparison between its differing service stacks.

The anonymous review bundle contains the fresh dataset, study plan, prompt packet, baseline predictions, projected C2 records, item-level final outputs, gold, analysis, redaction receipt, and frozen-manifest correction overlay. Its validator recomputes the complete statistics and both registered lane dispositions. The bundle also retains all 14 pilot attempts and both full passes from the adaptive extension, plus the original and public-control evidence and executable and Lean replay inputs. Searchable repository provenance is replaced with role descriptions. The private construction seed is excluded. Any change to labels, policy, prompts, controlled factors, or hidden-test bytes creates a new benchmark version. Bundle construction records the actual file count in its manifest, and validation recomputes that count from the generated tree.

9 AI Assistance Disclosure

Generative AI tools were used as assistants during literature search, research planning and methodological discussion, synthetic benchmark-template and code drafting and debugging, analysis checks and result interpretation, and parts of manuscript drafting and editing. The author chose the research question, defined the benchmark policy and experimental protocol, made the final study and analysis decisions, checked the generated benchmark through deterministic validators, checked the reported results against the released artifacts, checked the references, and is responsible for the final manuscript and submission. The quantitative results reported in this paper come from the validated study artifacts. The author retains authorship and final responsibility.

10 Conclusion

ClaimPromoteBench evaluates a narrow capability: selecting an evidence-state transition that conforms to a fixed author-defined policy on synthetic controlled pairs. The fresh hidden r6 result combines .9067 decision coverage with .8600 end-to-end item accuracy and .8200 pair coverage with .7533 end-to-end pair accuracy; its conditional values are .9485 and .9187. The public action-balanced challenge yields .7333/.5333 end-to-end item/pair accuracy for Qwen and .5000/.1444 for Flash, while `DEMOTE` recall remains 4/30 and 0/30. These results support service-configuration-specific policy conformance, not general research-agent competence.

The evidence chain also limits interpretation. All unusable responses remain in the denominator; the old Flash 401 lane is an interface failure, the newer route establishes executability only, and cross-stack outcomes are not a causal ablation. The deterministic executor is a definitional policy ceiling, whereas the Naive Bayes comparator is weak. The most important next tests are source-grounded naturalistic cases, stronger learned baselines, repeated service configurations, and independent assessment of whether the policy itself matches scientific practice.

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